

Fractional Dimensions — Beyond Whole Numbers

Name: _____ Date: _____ Class: _____

Build, observe, calculate, and explain.

Geometry World uses the familiar building and exploration of block-based 3D sandbox games to make geometric measurement visible. Count cubes in a layer, connect layers to multiplication, and compare strategies. Explaining and checking those calculations provides practice toward accurate, flexible calculation fluency.

What happens when dimensions aren't whole numbers? Explore how half-blocks change volume calculations.

Lesson goals

1. Compare a $4 \times 3 \times 2$ prism to a $4 \times 3 \times 2.5$ prism
2. Calculate volume when one dimension is a fraction
3. Understand that volume can be a non-whole number
4. Estimate the volume of a partially filled container

Your lesson route

1. Fraction Prof · investigation
2. Between Quiz · 3 question steps
3. Container Challenge · 2 question steps
4. Height Detective · 3 question steps

Print in portrait at 100% or Save as PDF. This companion follows the current lesson's tasks and question choices. Answer by hand; a drawing, words, or another clear representation can show your thinking. Use the world to gather observations before completing the paper independently.

Fraction Prof

Guide: **Fraction Prof** at world coordinates (3, 3, 3). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

In real life, dimensions aren't always whole numbers! A box might be $4 \times 3 \times 2.5$ feet. Volume = $4 \times 3 \times 2.5 = 30$ cubic feet. The formula still works!

Self-review: Check the task requirements, label units, and explain how your evidence supports your result.

Plan / sketch · label dimensions and units

Measure / calculate · record the quantities and the operations you choose

Explain and review

How does your evidence support your result? Check it with a second method or describe a useful revision.

My mathematical reasoning and revisions

Between Quiz · Questions 1–2

Guide: **Between Quiz** at world coordinates (12, 5, 3). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 1 · Step 1

Blue is $4 \times 3 \times 2 = 24$. Gold is $4 \times 3 \times 4 = 48$. Halfway height is?

- A. 3.5 B. 2.5 (halfway between 2 and 4-1=3) C. 3

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Question 2 · Step 2

Volume of $4 \times 3 \times 2.5$?

- A. 24 cubic units B. 36 cubic units C. 30 cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Between Quiz · Questions 3

Guide: **Between Quiz** at world coordinates (12, 5, 3). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 3 · Step 3

Is 30 between 24 and 48?

A. Yes! Fractional height gives in-between volume B. No C. Only sometimes

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Container Challenge · Questions 4–5

Guide: **Container Challenge** at world coordinates (4, 1, 9). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 4 · Step 1

Sand volume? ($4 \times 3 \times 2$)

- A. 36 cubic units B. 24 cubic units C. 12 cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Question 5 · Step 2

Total container = $4 \times 3 \times 3 = 36$. Empty space = $36 - 24 = ?$

- A. 24 more B. 6 more C. 12 more cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Height Detective · Questions 6–7

Guide: **Height Detective** at world coordinates (17, 1, 9). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 6 · Step 1

Short tower volume? ($3 \times 3 \times 3$)

- A. 27 cubic units B. 18 cubic units C. 9 cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Question 7 · Step 2

Tall tower? ($3 \times 3 \times 5$)

- A. 30 cubic units B. 45 cubic units C. 27 cubic units

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Height Detective · Questions 8

Guide: **Height Detective** at world coordinates (17, 1, 9). Use this sheet alongside the lesson world. Observe the named models or guide before answering; sketch and label the model so your reasoning can be understood on paper.

Question 8 · Step 3

$45 \div 27 \approx ?$ (how many times bigger?)

A. Exactly $2\times$ B. About $1.5\times$ C. About $1.67\times$ ($5/3$)

Sketch / label the model, or describe what you observe

Calculations · show an equation, units, and intermediate steps

Reasoning · explain your strategy and check your answer

My answer (include units): _____

Connect the ideas

Choose one model or problem from this lesson. Explain how its representation connects to your calculations.

Model / diagram · show cubes, layers, dimensions, or a decomposition that fits the task

First calculation strategy · show the steps and units

Check with another strategy · explain why it agrees or what you revised

Reflection · what became easier and what do you still want to investigate?